An Introduction to Deep (Machine) Learning

For years, humans have tried to get computers to replicate the thinking processes of the human brain. To a limited extent, this has been possible by using deep learning and deep neural networks. This article provides insights into deep or machine learning.

When surfing the Net or while on social media, you must have wondered how you automatically get pop-ups of things that interest you. Did you know that there are lots of things happening behind the scenes? In fact, lots of computations and algorithms are running in the background to automatically find and display things that interest you, based on your search history… And this is where deep learning begins.

Deep learning is one of the hottest topics nowadays. If you do a Google search, you will come across a lot that's happening in this field and it is getting better every day, as one can gauge when reading up on, 'Artificial intelligence: Google's AlphaGo beats Go master Lee Se-dol' at *www.bbc.com.*

In this article, we will look at how we can practically implement a three-layer network for deep learning and the basics to understand the network.

Definition

It all started with machine learning – a process by which we humans wanted to train machines to learn as we do. Deep learning is one of the ways of moving machine learning closer to its original goal—artificial intelligence.

As we are dealing with computers here, inputs to these are data such as images, sound and text. Problem statements include image recognition, speech recognition, and so on. We will focus on the image recognition problem here.

History

When humans invented computers, scientists started working on machine learning by defining the properties of objects. For instance, the image of a cup was defined as cylindrical and semi-circular objects placed close to each other. But in this universe, there are so many objects and many of them have similar properties. So expertise was needed in each field to define the properties of the objects. This seemed to be a logically incorrect method as its complexity would undoubtedly increase with an increase in the number of objects.

This triggered new ways of machine learning whereby machines became capable of learning by themselves, which in turn led to deep learning.

Architecture

This is a new area of research and there have been many architectures proposed till now. These are:

1. Deep neural networks
2. Deep belief networks
3. Convolutional neural networks
4. Convolutional deep belief networks
5. Large memory storage and retrieval (LAMSTAR) neural networks
6. Deep stacking networks